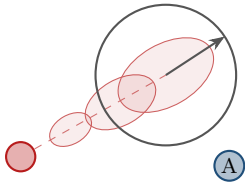
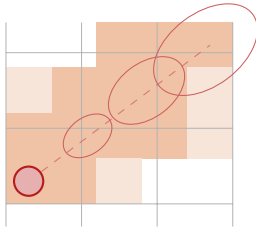


predicted $\hat{\mathbf{p}}_{T+h}$ with Σ_h



disk radius $r_A + r_B + k_p \sqrt{\lambda_{\max}(\Sigma_h)}$

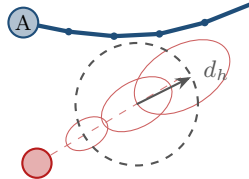
(a) ORCA: inflated disk



cells inside $\mathcal{E}_h(p)$ get cost c_{obs} in time layer h

(b) D* Lite: cost region

$\|\mathbf{p}_{A,h} - \hat{\mathbf{p}}_{T+h}\| \geq d_h$ for each h



$d_h = r_A + r_B + k_{1-\delta} \sqrt{\lambda_{\max}(\Sigma_h)}$

(c) MPC: chance constraint